North Carolina State University

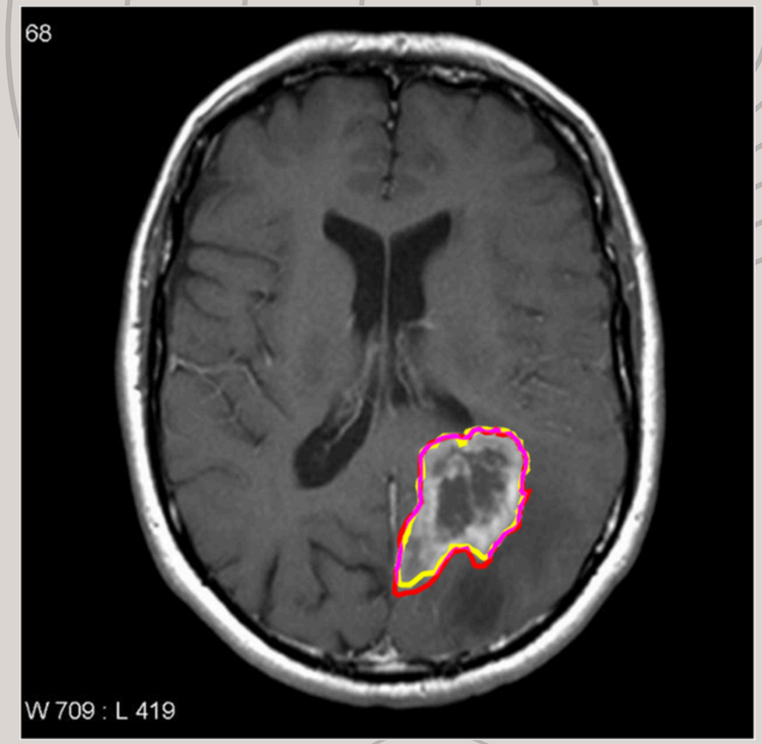
Brain Tumor Detection Using Deep Learning

Presented by Group 154

Dhairya Shah
Akshat Shah
Gaurav Mehta

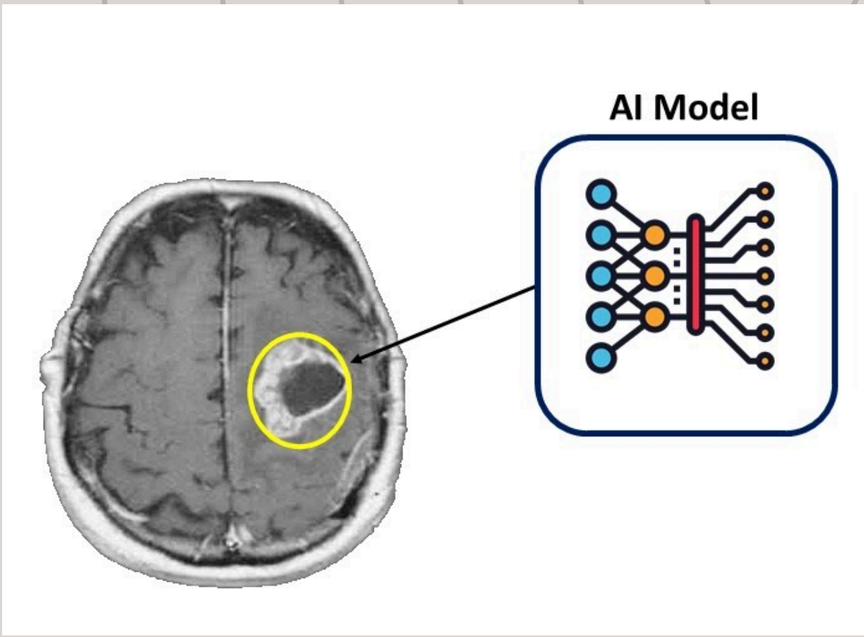
Problem Statement

- This project aims to facilitate automation tumor detection and classification in MR images using deep learning techniques, enhancing diagnostic accuracy and efficiency.
- Convolutional Neural Networks (CNNs) are utilized for feature extraction, segmentation, and classification of brain tumors.



Motivation

- Enhanced Early Detection: CNNs improve early brain tumor detection, leading to better treatment outcomes and higher survival rates.
- Reduced Human Error: Automating tumor identification with CNNs minimizes diagnostic errors and supports healthcare professionals in accurate diagnosis.



Dataset Description

The dataset consists of 4 major categories of images:

- Glioma: A tumor that arises from glial cells in the brain, often aggressive and difficult to treat.
- Meningioma: A slow-growing tumor that forms in the meninges, the protective layers surrounding the brain .
- Pituitary Tumor: A usually benign tumor in the pituitary gland that can affect hormone production and bodily functions.
- No Tumor: MRI images showing normal brain tissue without any detectable tumors.

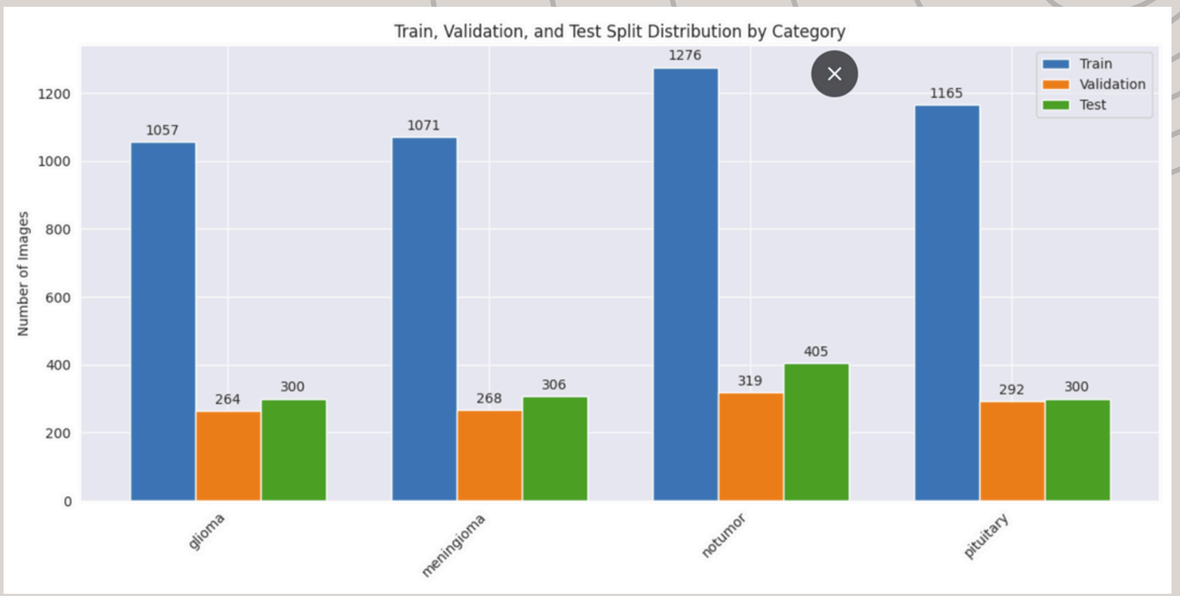
Dataset Description

Dataset is taken from Kaggle(Open Source)

Link:https:www.kaggle.com/datasets/masoudnickparvar/brain-tumor-mri-dataset

Number of images for each category

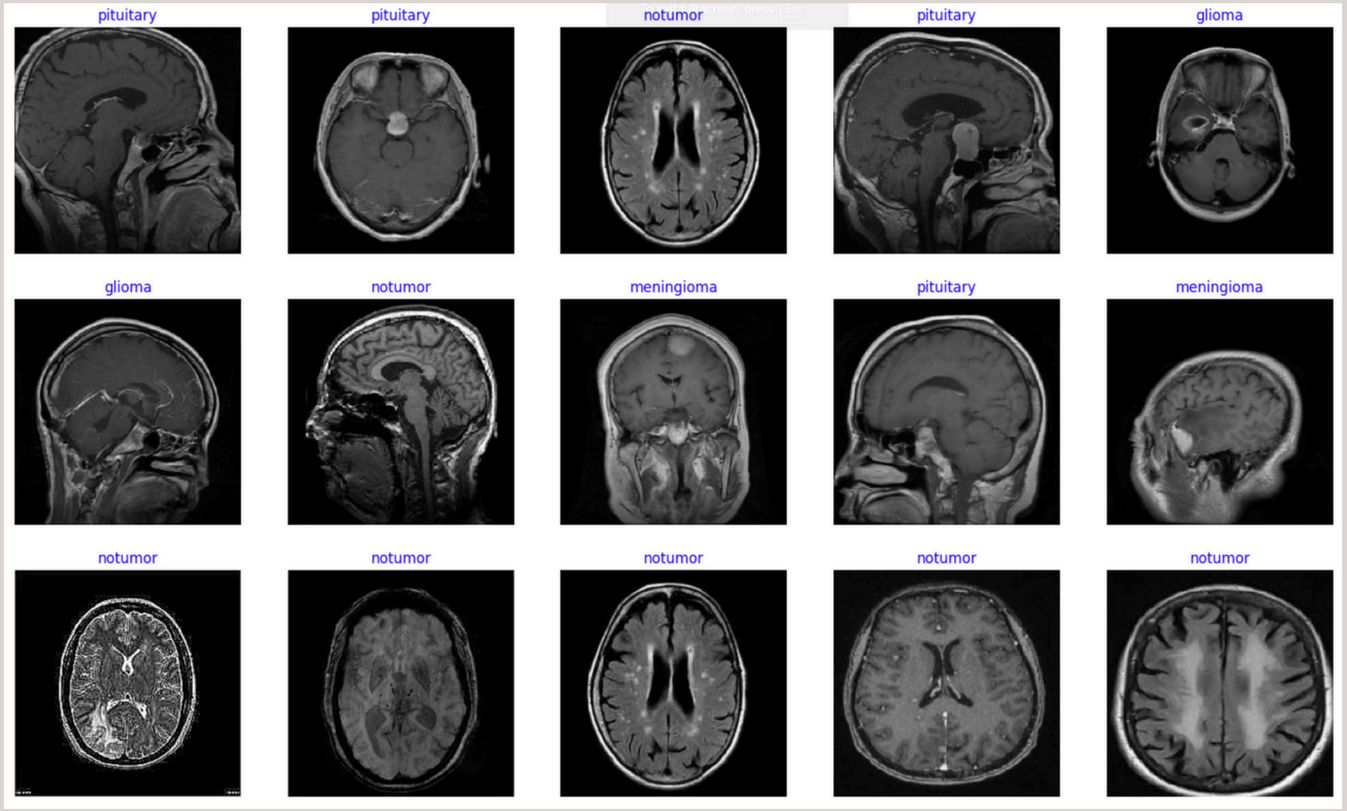
- glioma = 1621
- meningioma = 1645
- no tumor = 2000
- pituitary = 1757



Preprocessing Steps performed:

- Converted all images to 224×224 to maintain uniform input size.
- Applied horizontal flipping to enhance model generalization (for training only).
- Converted labels into a categorical format for multi-class classification, using one hot encoding.

Input - Labeled Images



Output - Correctly Categorized data

Baseline Model

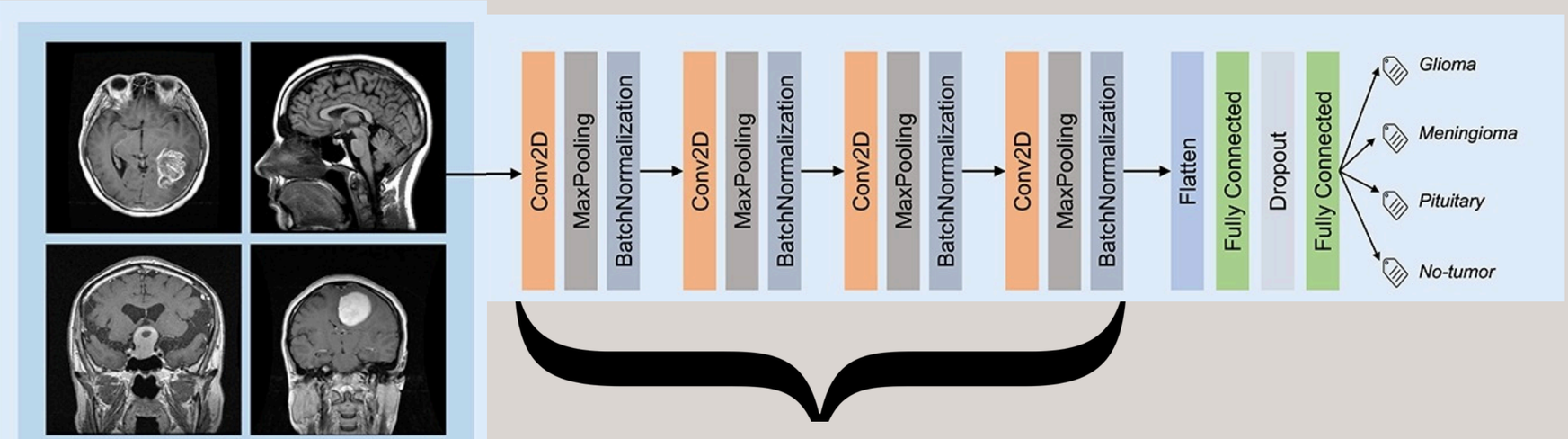
- We implemented a standard Convolutional Neural Network (CNN) framework consisting of sequential Conv2D layers, MaxPooling, and Batch Normalization. This combination allows the model to automatically learn and extract low-to-high-level spatial features from input images.
- The extracted features are flattened and passed through Dense layers, with Dropout added to prevent overfitting. The final Softmax layer outputs class probabilities, making the architecture suitable for generic image classification tasks.

Model: "sequential"

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 224, 224, 32)	896
max_pooling2d (MaxPooling2D)	(None, 112, 112, 32)	0
batch_normalization (BatchNormalization)	(None, 112, 112, 32)	128
conv2d_1 (Conv2D)	(None, 112, 112, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(None, 56, 56, 64)	0
batch_normalization_1 (BatchNormalization)	(None, 56, 56, 64)	256
conv2d_2 (Conv2D)	(None, 56, 56, 128)	73,856
max_pooling2d_2 (MaxPooling2D)	(None, 28, 28, 128)	0
batch_normalization_2 (BatchNormalization)	(None, 28, 28, 128)	512
flatten (Flatten)	(None, 100352)	0
dense (Dense)	(None, 128)	12,845,184
dropout (Dropout)	(None, 128)	0
dense_1 (Dense)	(None, 4)	516

Total params: 12,939,844 (49.36 MB)
Trainable params: 12,939,396 (49.36 MB)
Non-trainable params: 448 (1.75 KB)

Proposed CNN Architecture



EfficientNet B3:

It is a lightweight, pre-trained convolutional neural network (CNN) model known for achieving high accuracy with fewer parameters by using a compound scaling method. It balances depth, width, and resolution for optimal performance in image classification tasks.

Hyper Parameters Tuning

Batch_size = 40 (Number of training samples used in one pass)

Epochs = 20 (Total number of times the model will iterate)

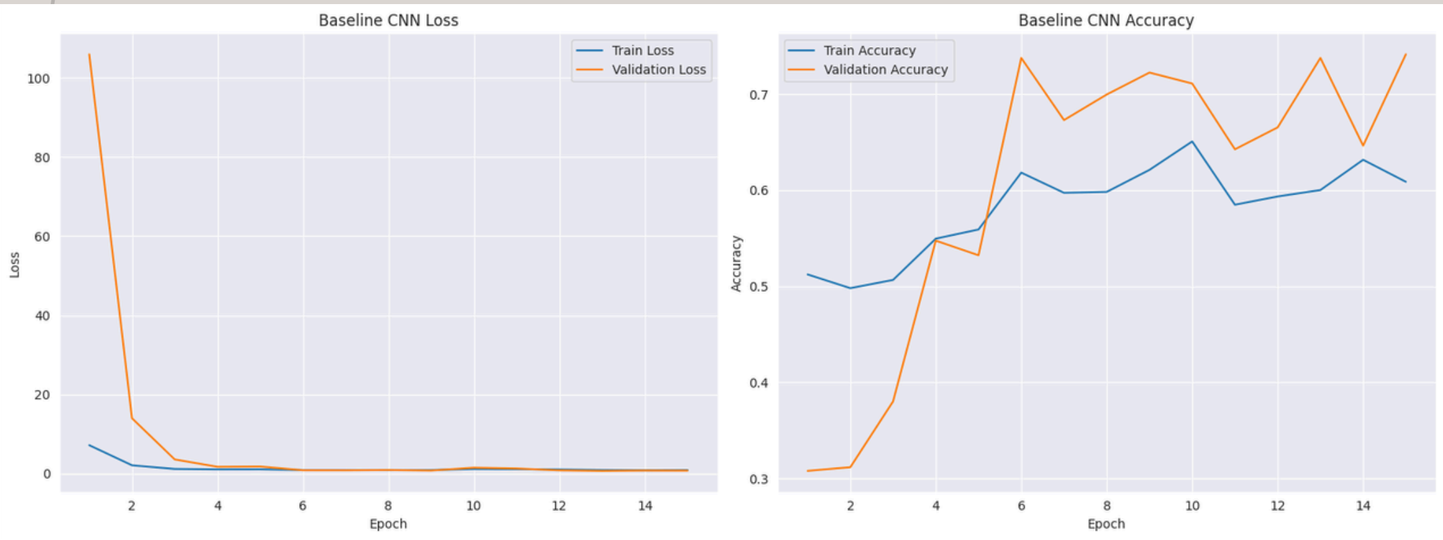
Factor = 0.7 (Multiplicative factor for reducing the learning rate)

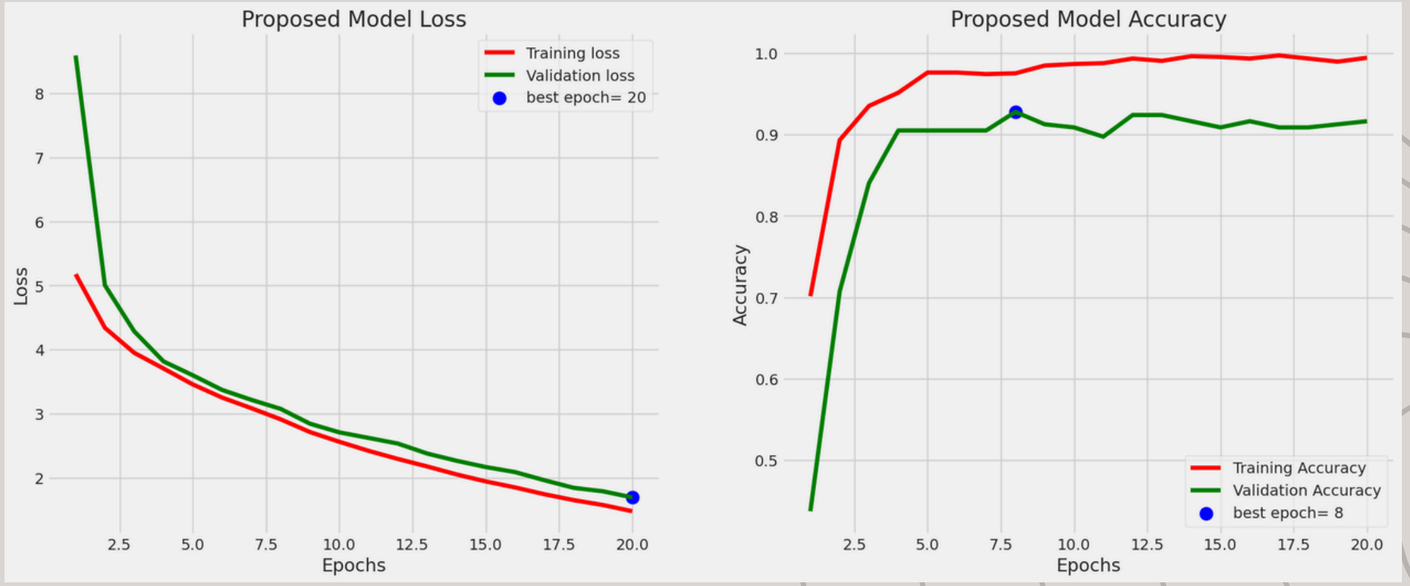
Ask_epoch = 5 (Use to check for Improvement after 5 epochs)

warmup = 3 (Defines how many initial epochs to skip before starting to apply learning rate)

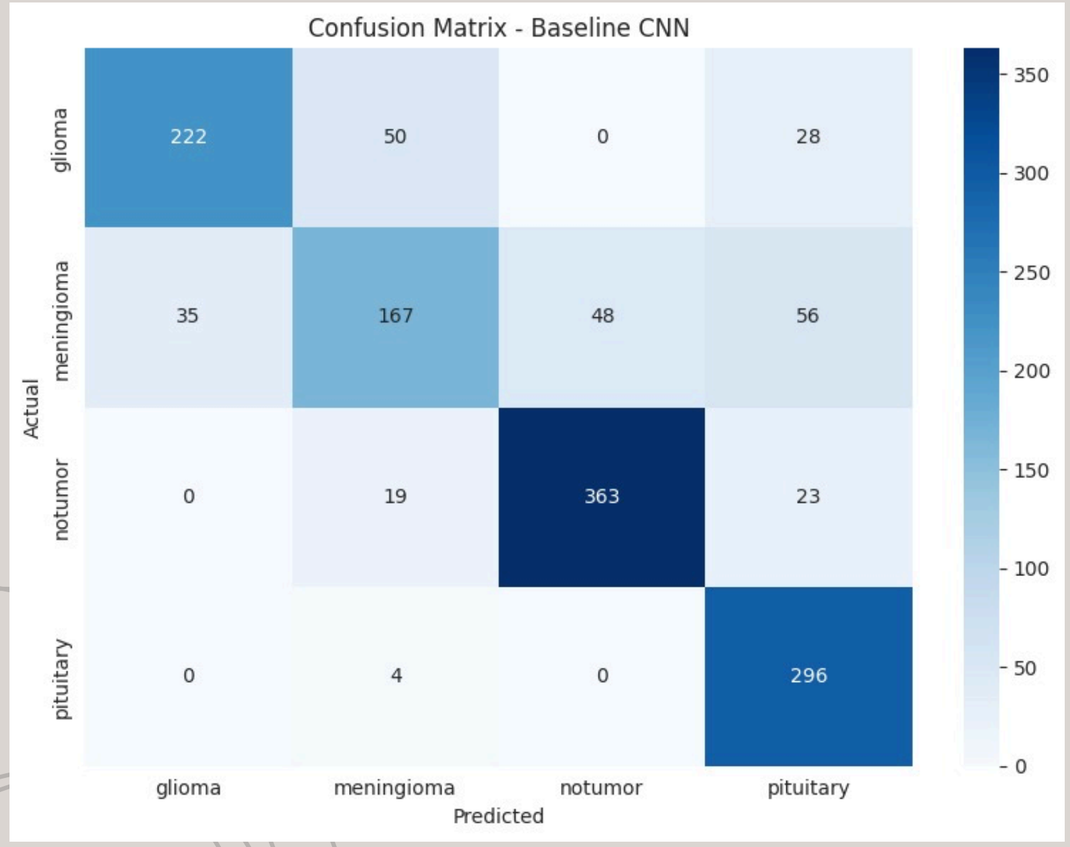
patience = 2 (Defines number of epochs to wait after the last improvement before reducing the learning rate.)

Comparison of Models

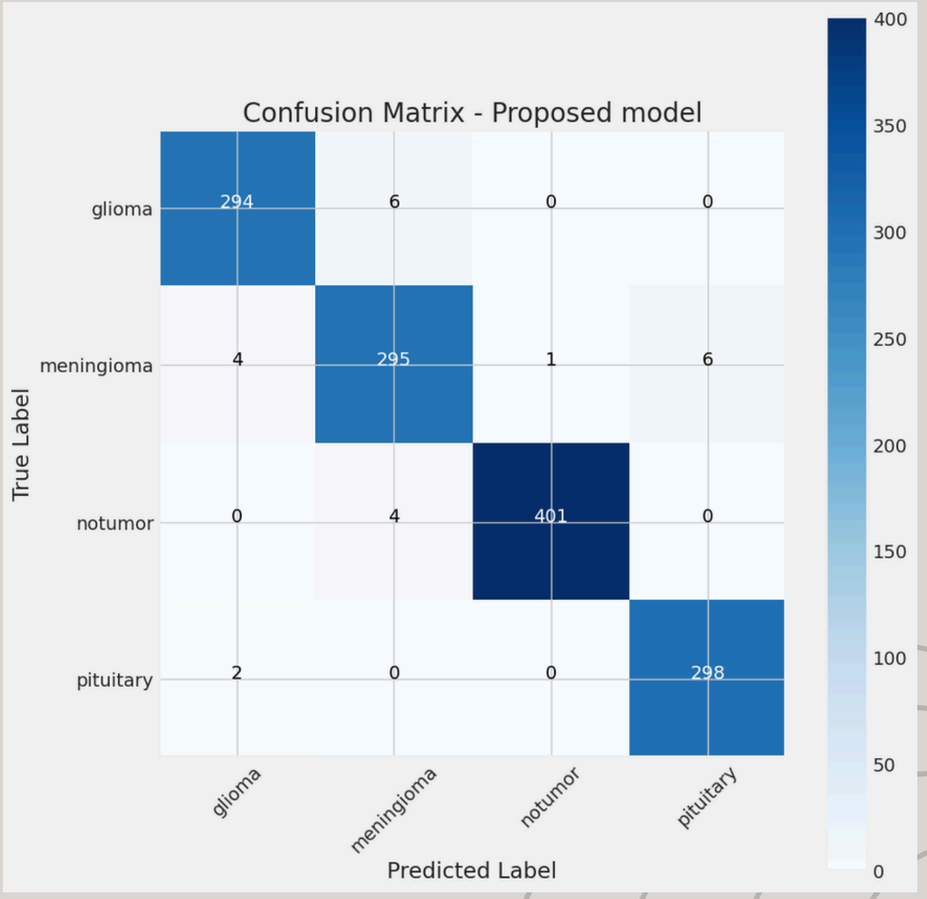




Comparison of Models



Baseline CNN - Test Accuracy: 79.94%



Proposed Model - Test Accuracy: 99.86%

Performance of Proposed Model

```
Train Loss:  1.4157873392105103
Train Accuracy:  0.9986842274665833
--------------------
Validation Loss:  1.6893080472946167
Validation Accuracy:  0.9163498282432556
--------------------
Test Loss:  1.470223069190979
Test Accuracy:  0.9824561476707458
```

	precision	recall	f1-score	support
glioma	0.98	0.98	0.98	300
meningioma	0.97	0.96	0.97	306
notumor	1.00	0.99	0.99	405
pituitary	0.98	0.99	0.99	300
accuracy			0.98	1311
macro avg	0.98	0.98	0.98	1311
weighted avg	0.98	0.98	0.98	1311

Prior Group Work

- This project builds on prior research in medical imaging and deep learning.
- Our group member does have previous project experience with the the CNN model particularly Gaurav.
- Dhairya has professional experience working on general ML Models
- Akshat has previously worked with CNN classifiers.
- NVIDIA DLI course completed by all the members of the team.

References

[1] Brain Tumor MRI Dataset – https://www.kaggle.com/datasets/masoudnickparvar/brain-tumor-mri-dataset
[2] Automation of Brain Tumor Identification using EfficientNet on Magnetic Resonance Images – https://www.sciencedirect.com/science/article/pii/S1877050923001333
[3] Automated brain tumor classification using various deep learning models: a comparative study – https://ijeecs.iaescore.com/index.php/IJEECS/article/view/22770

North Carolina State University

Thank You